H

NATIONAL JUNIOR COLLEGE

Higher 2

CANDIDATE NAME

BIOLOGY CLASS

REGISTRATION NO.

BIOLOGY 9747/02

JC2 Preliminary Examinations
Paper 2

22 Sept 2008

2 Hours

READ THESE INSTRUCTIONS FIRST

Write your name and registration number (e.g. 05IP0409 or 07S0620) and Biology Class (e.g. 2bi2E) clearly on the cover page.

Write your Name, Registration No. and Biology Class on the writing paper that you write the essay on.

Write in dark blue or black pen.

Answer **all** questions in **Section A** and **choose one** essay question to answer from **Section B**.

At the end of the exam, staple your essay to the back of the question paper.
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINERS' USE ONLY	
1	
2	
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8 / 9	
TOTAL (/100)	

This paper consists of **18** printed pages including cover page.

SECTION A: 80 MARKS

Answer all questions.

Question 1

Gastric proton/potassium-ATPase (H^+/K-ATPase) is the major enzyme involved in gastric acid secretion in the stomach. Recognizing the calming effects that tea can have on an upset stomach, biochemists quickly showed that the tannic acid found in tea is a potent inhibitor of this enzyme. Gastric proton/potassium-ATPase (H^+/K-ATPase) has two substrates, potassium ion (K^+) and ATP. (K_m is substrate concentration at ½ V_{max})

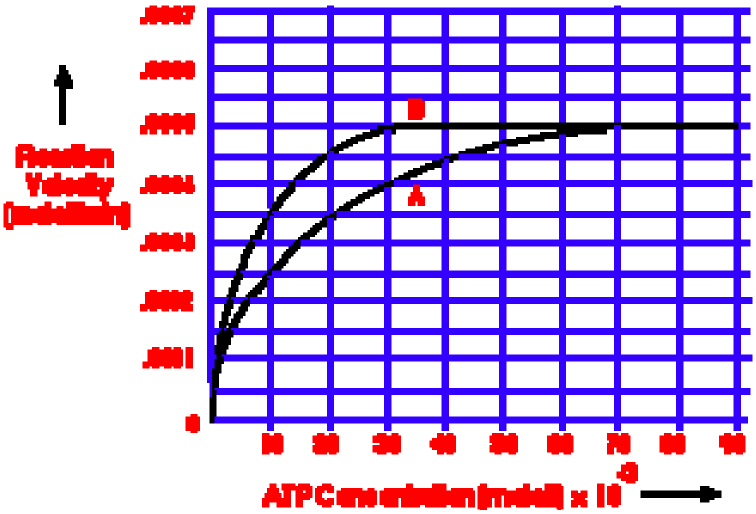


Figure 1.1 Reaction velocity-substrate concentration curves for Gastric proton/potassium-ATPase (H^+/K-ATPase) with unlimiting levels of K^+ but varying levels of ATP. **A:** *with tannic acid;* **B:** *without tannic acid.*

(a) What is the minimum ATP concentration needed to saturate the enzyme in this system without tannic acid?

____

____ [1]

(b) By comparing K_m values based on the graphs in figure 1, explain which treatment would result in the higher affinity of the enzyme for the substrate?

____

____

____ [2]

(c) i) Name the type of inhibition that is involved above and provide evidences for your choice.

[2]

ii)With reference to the structure of the active site, suggest the possible mechanism of inhibition and why the inhibitor cannot be broken down by the enzyme.

[2]

(d) Describe two ways in which enzymes interact with substrates to specifically increase the likelihood of a reaction.

[2]

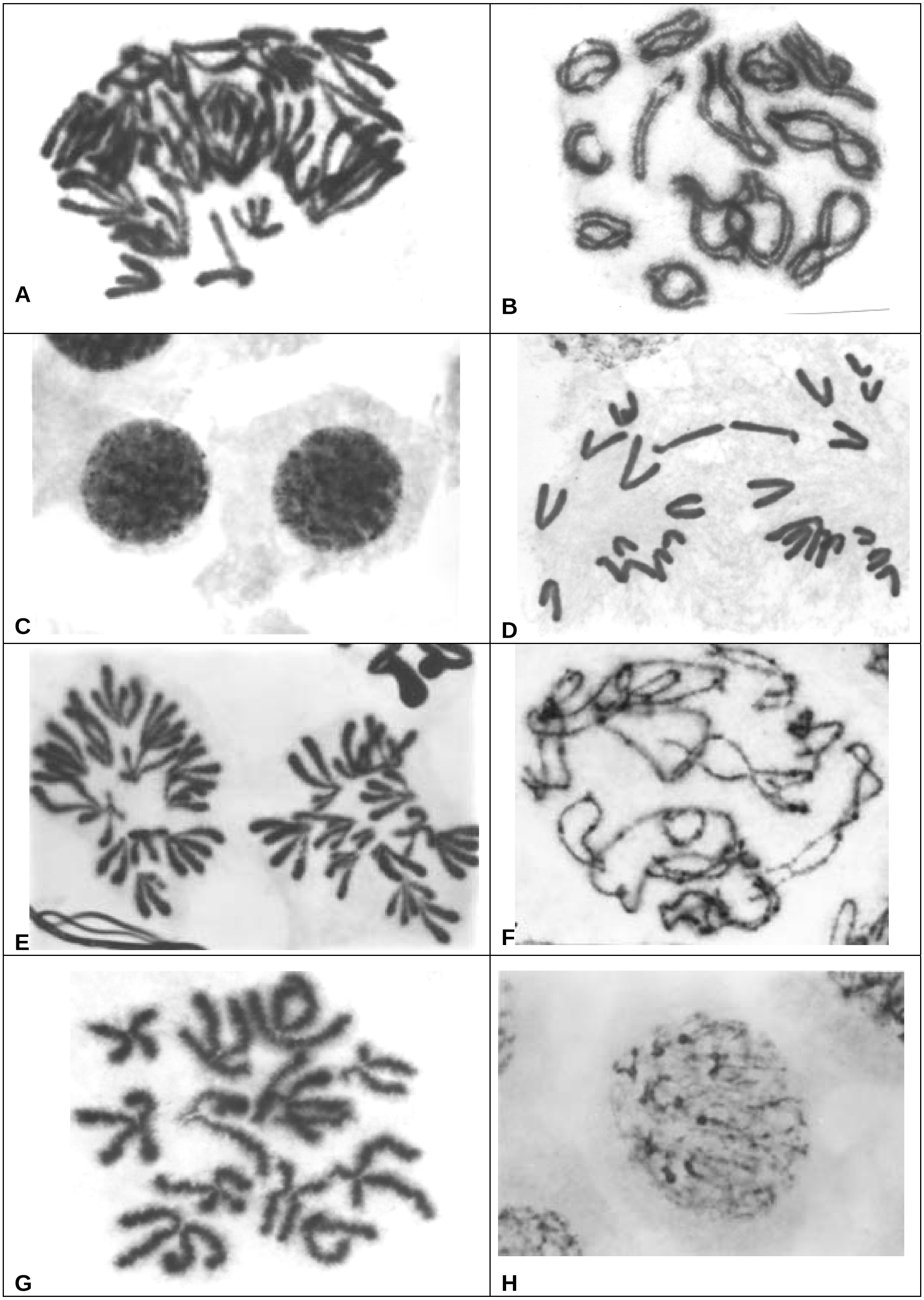
(e) Giving an example in each, describe one main difference between prosthetic groups and coenzymes that are involved in enzymatic reactions.

[2]

[Total: 11 marks]

Question 2

The micrographs below show nuclei of cells at various stages during nuclear division in a flowering plant.



(a) Micrographs **C** and **H** show the 5th and 6th stages in the sequence. In the boxes below, arrange the other letters shown on the micrographs to indicate the correct chronological sequence for this type of nuclear division.

				C	**H**		

[2]

(b) What is the diploid number of this plant? With reference to at least 2 stages from the micrographs, explain how you arrive at your answer.

[3]

(c) Discuss the significance of stage **B**.

[3]

[Total : 8 marks]

Question 3

The diagram below (Fig. 3.1) shows the process of transcription of DNA to form RNA at a particular point in time.

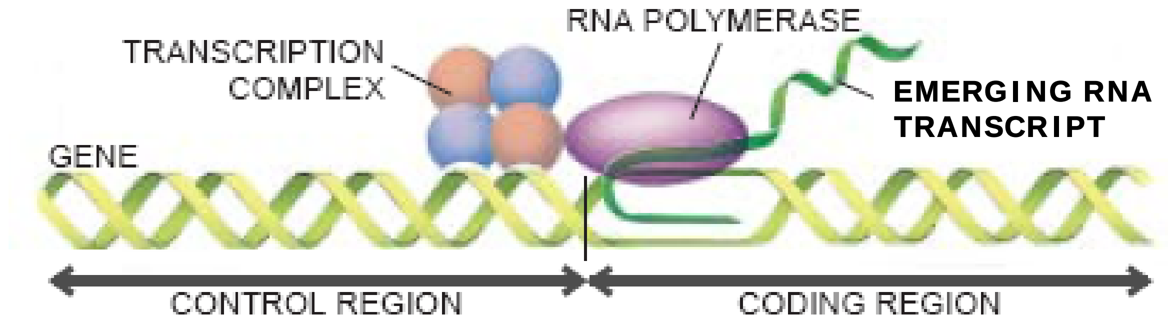


Diagram from Sci Am. Dec 1994

Fig. 3.1

(a) i) What are the main characteristics in the structure of DNA?

____

____

____

____

____

____

____

____ [4]

ii) Describe the events that will occur after the point in time shown in Fig. 3.1 until the completion of transcription.

____

____

____

____

____ [3]

These subsequent diagrams ***b*** and ***c*** in Fig. 3.2 and Fig. 3.3 show what happens when an artificially synthesized oligonucleotide binds to the DNA. This is a method to silence the expression of the gene.

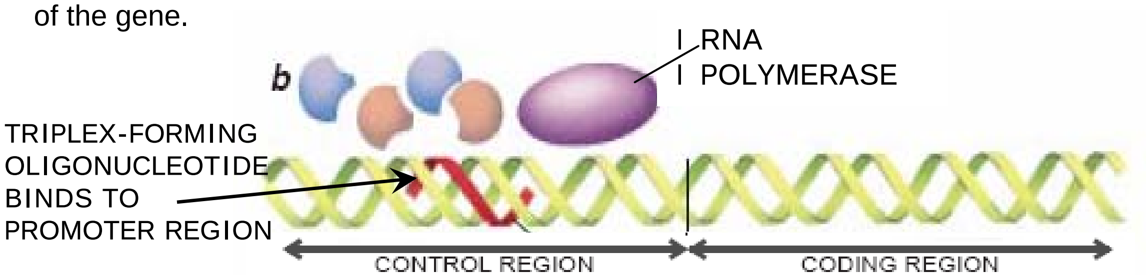


Diagram from Sci Am. Dec 1994

Fig. 3.2

(b) i) Fig.3.2 above (Diagram ***b***) shows the artificially synthesized oligonucleotide binding to the major groove of the DNA, thus forming a triple helix (triplex) in that particular region where it binds.
Explain why, according to Fig. 3.2, the RNA polymerase cannot bind to the DNA at all.

[2]

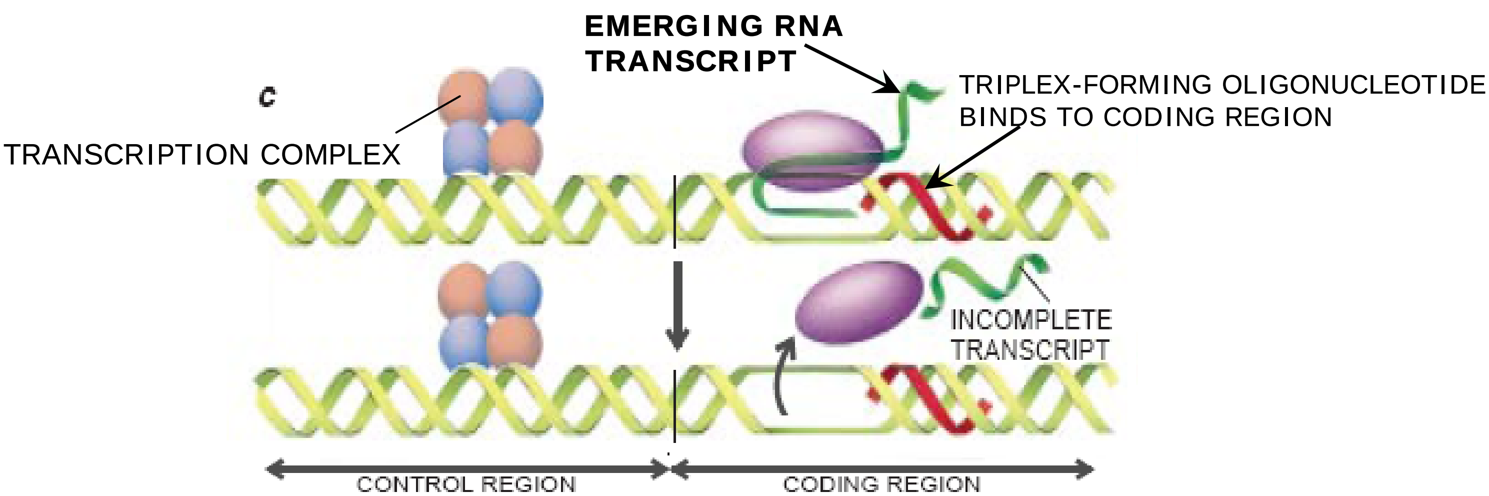


Fig. 3.3 *Diagram from Sci Am. Dec 1994*

ii) Diagram ***c*** above shows the scenario whereby the artificially synthesized oligonucleotide binds to the coding region of the DNA. With reference to Fig. 3.2 and Fig. 3.3, account for the difference between the outcomes of Diagram ***b*** and ***c***.

[3]

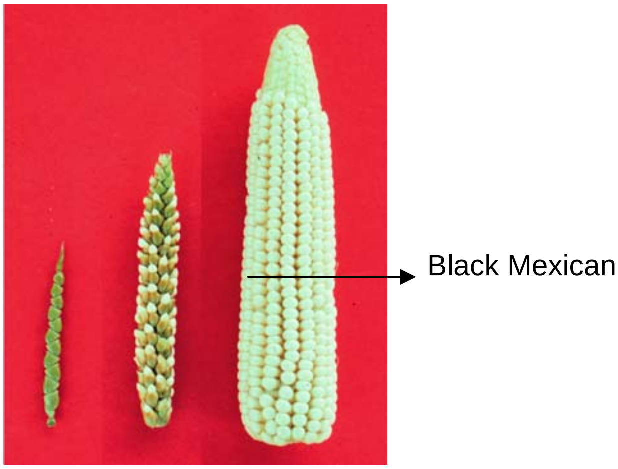
(c) Despite the differences between the outcomes of Diagram b and c, both eventually silence the expression of the gene.
Thus briefly explain what it means to "silence the expression of the gene".

[1]

[Total: 13 marks]

Question 4

The diagram below shows the ears of maize *(Zea mays*). On the left, is variety Tom Thumb, the Black Mexican one on the right, and that of their F1 hybrid in the centre



(photo by John Doebley)

A cross was made between 2 pure breeding maize varieties, Tom Thumb and Black Mexican which differed markedly in ear length. The ear length of both parents, the F1 and the F2 generations, was measured in centimetre. This is given in the table below with the number of ears in each length category (e.g.14 of the F1 plants produced ears 12 cm in length).

	Ear length (cm)																
	5	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**
Black Mexican parents									3	11	12	15	26	15	10	7	2
Tom Thumb parents	4	21	24	8													
F1					1	12	12	14	17	9	4						
F2			1	10	19	26	47	73	68	68	39	25	15	9	1		

(a) Both parents showed variations in ear length. The variations in the F1 is comparable to the average of the parental variations while the variations in the F2 is greater than that found within the parental lines or the F1.

i) Suggest an explanation for this.

[3]

Part (a) continued…

ii) Explain the genetic basis behind the wide range of ear lengths.

[2]

iii) What is the term used to describe the range of phenotypes in the above example?

[1]

(b) In another experiment, pea plants with round, black seeds were crossed with pure-breeding pea plants with wrinkled, green seeds. The offspring all had round, black seeds. These seeds were grown and the resultant plants allowed to self-pollinate (fertilise).

i) What would be the expected phenotypic ratio? Using appropriate symbols draw a genetic diagram to explain your results. [5]

The above cross in (b)(i) produced 324 offspring with the following characteristics.

Number of offspring	Phenotype
181	round, black seeds
51	round, green seeds
53	wrinkled, black seeds
39	wrinkled, green seeds

ii) Complete the table below and calculate the chi-square to test the significance of the differences between the observed and expected results. [2]

	round, black seeds	round, green seeds	wrinkled, black seeds	wrinkled, green seeds
Observed number (O)	181	51	53	39
Expected number (E)				
Difference between observed & expected numbers (O – E)				
$(O – E)^2$				
$(O – E)^2 / E$				
$\chi^2 = \Sigma [(O – E)^2 / E]$				

	Probability, p				
Degrees of freedom	**0.10**	**0.05**	**0.02**	**0.01**	**0.001**
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

iii) With reference to the table of probabilities (above), explain how the value for the chi-squared test supports the hypothesis that these are two pairs of segregating alleles at two loci.

[2]

[Total: 15 marks]

Question 5

(a) i) State two random processes that contribute to the accumulation of allele, which do not have any benefit to survival value in a population.

[2]

ii) Explain the basis behind the accumulation of such genes.

[1]

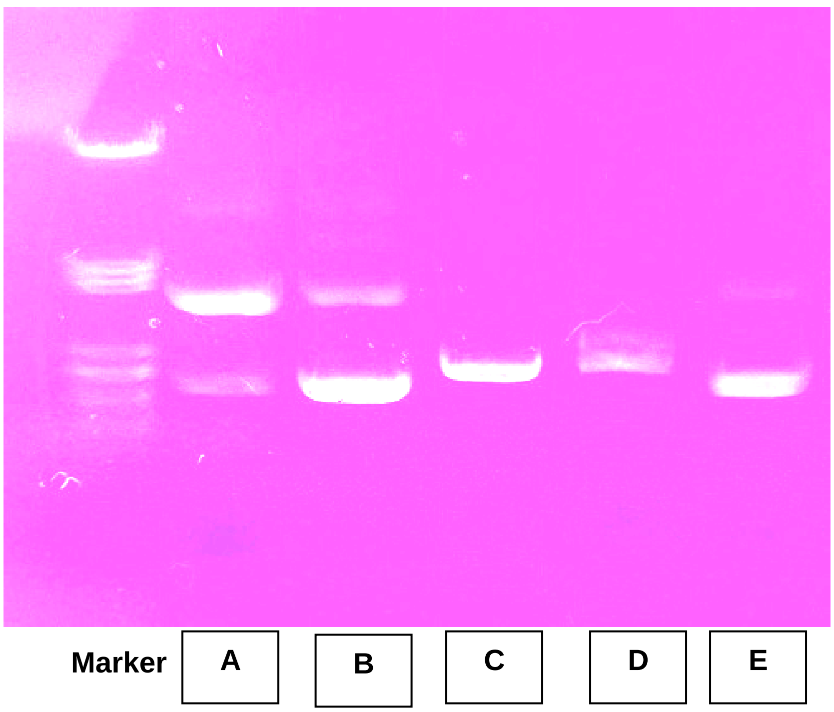


Fig. 5.1 shows a gel electrophoresis result of DNA from a gene, obtained from similar looking flagella obtained from 5 species of marine bacteria. The DNA samples were treated with the same restriction enzyme. Species A to E were each retrieved from different oceans of the world.

(b) i) Name the two species that are most likely to evolve flagella similar to A, as a result of convergent evolution.

[1]

ii) Explain how the method of analysis has supported your choice of answer.

[2]

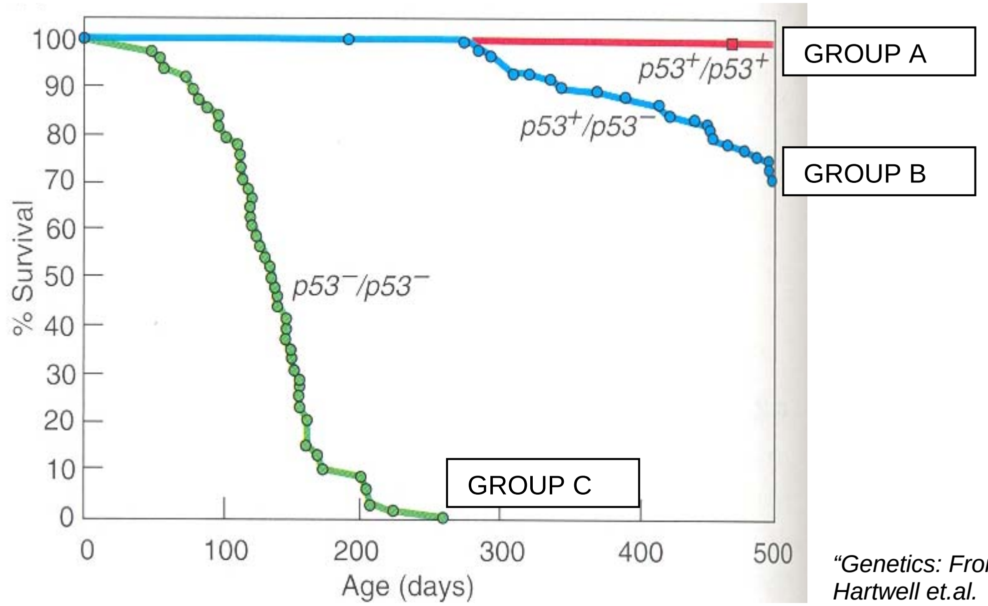
(c) Give four advantages of using such a method in the classification of the 5 organisms into the phylogenetic tree.

[4]

[Total: 10 marks]

Question 6

The graph below (Fig. 6.1) shows the percentage survival of mice with different genotypes at the p53 gene locus. $p53^+$ is the wild-type allele, and $p53^-$ is the mutant allele.



"Genetics: From Genes to Genomes" Hartwell et.al.

Fig. 6.1

(a) Account for the difference between Group B and Group C mice.

____

____

____ [2]

(b) Filling up the table below, contrast the possible genetic changes that may lead to tumour formation in the p53 gene and in a proto-oncogene. [3]

	Possible changes to p53 wild type allele	**Possible changes to proto-oncogene**
Point mutation in gene itself leads to:		
Point mutation in control element leads to:		
Gene amplification:		
Translocation of gene to:		

(c) i) Describe the significant changes in a mouse cell from the point of acquiring of the mutant p53 alleles to the cell becoming cancerous.

[4]

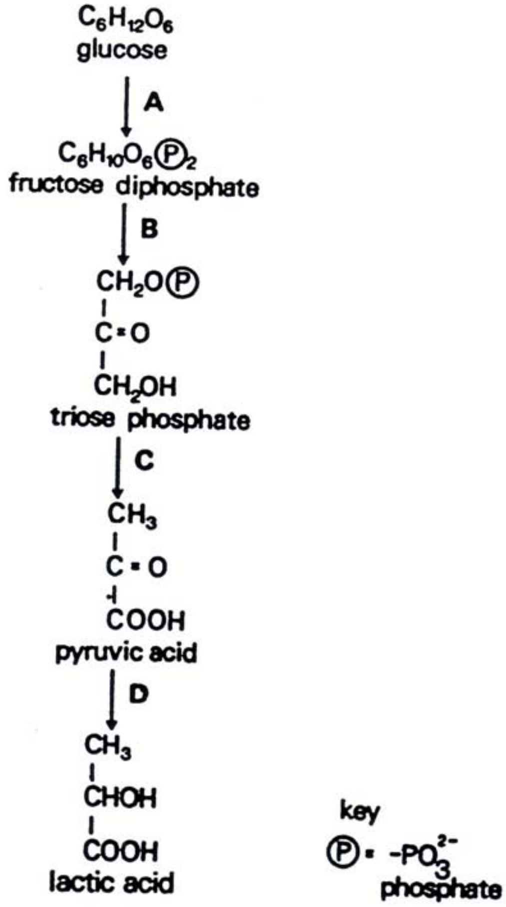
ii) How can one cancerous cell eventually lead to the death of the mouse?

[3]

[Total: 12 marks]

Question 7

In anaerobic respiration in muscle tissue the conversion of glucose to lactic acid can be represented as a four stage process as follows:



(a) Name the four stages in terms of the chemical processes occurring.

A ____________________

B ____________________

C ____________________

D ____________________ [2]

(b) At what stage would you expect hydrogen transfer to NAD to occur?

____________________ [1]

(c) Is there a net change in the concentrations of NAD and $NADH_2$ in the whole process? Explain your answer.

[3]

(d) If one molecule of glucose is converted to two molecules of lactic acid, what is the net synthesis of ATP? Explain your answer.

[1]

Explanation:

[2]

(e) In what ways is the ethanol formation different from that of lactate?

[2]

[Total: 11 marks]

SECTION B: 20 MARKS

Choose **EITHER** Question 8 **OR** Question 9.

Write your answers to this question on a separate piece of writing paper.
Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections (a), (b), etc., as indicated in the question.

Question 8

a) Compare the structure and organization of prokaryotic and eukaryotic genomes. [8]

b) Describe the various ways in which the activity of an enzyme may be controlled after a mature mRNA that codes for it has been produced in a eukaryotic cell. [6]

c) Discuss the ways in which DNA is normally transferred from one bacterium to another without any human interference. [6]

Question 9

a) With reference to the three different stages of cell signaling, describe in detail the sequence of events when a glucagon molecule reaches the liver cell. [10]

b) Explain how a strong stimuli can be transmitted from the presynaptic membrane of a sensory neuron to an internuncial neuron. [7]

c) Explain how loss of myelination, which happens in the demyelinating disease multiple sclerosis, disrupts signal transmission in the nervous system. [3]

--- The End ---